Particle Trajectory Representation Learning with Masked Point Modeling

Point-based Liquid Argon Masked Autoencoder (PoLAr-MAE)

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Liquid Argon Time Projection Chambers

When charged particles traverse the argon, they leave trails of ionization electrons. These electrons drift under a strong electric field and are collected on finely segmented anode planes, allowing for precise reconstruction of particle paths and energy depositions, often achieving millimeter resolution over meter-scale distances (visualized in Fig. 1).

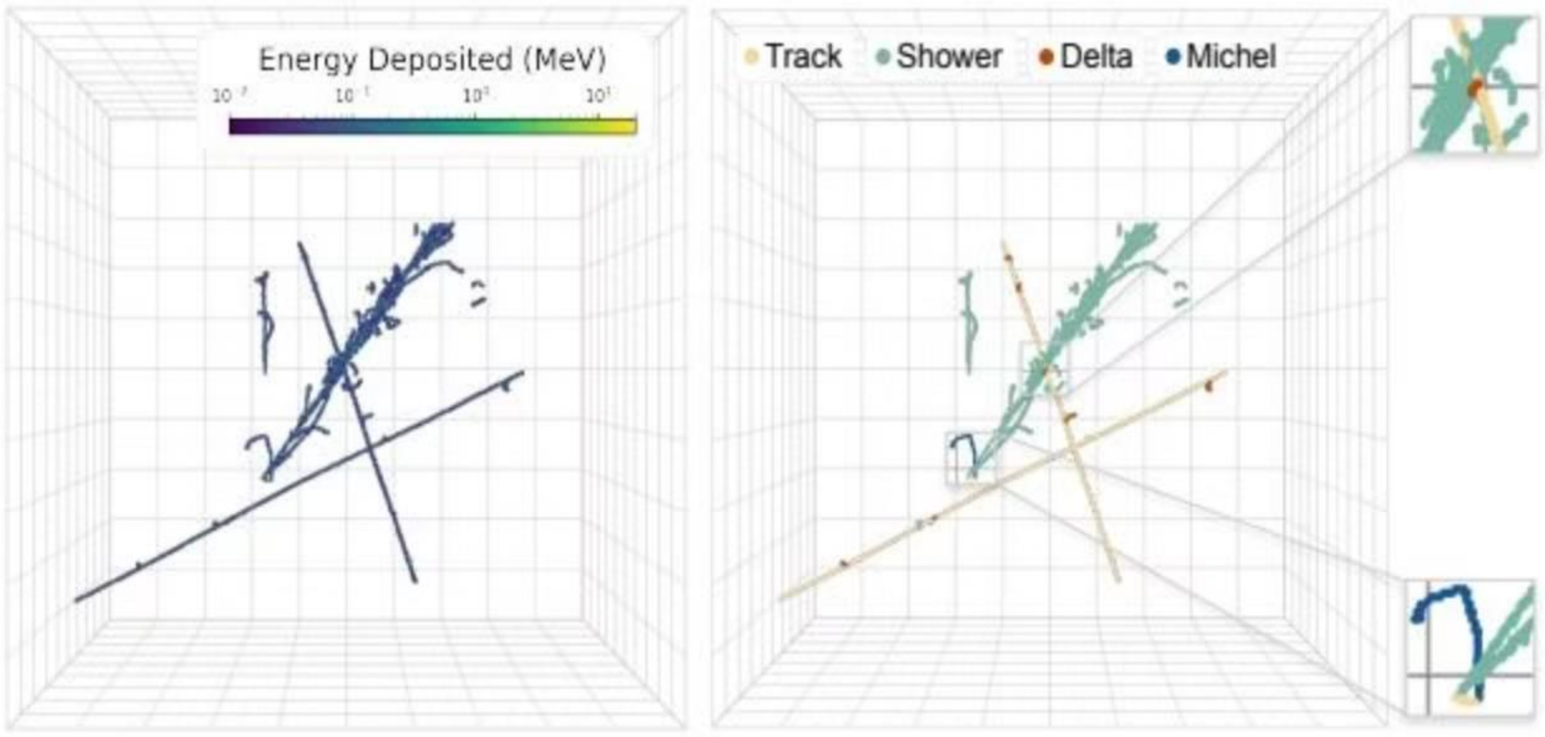

Figure 1: LArTPC Data Illustration

Track-like Particles

Minimum ionizing particles like muons, protons, and charged pions produce long, relatively straight tracks characterized by dense, linear ionization trails.

Electromagnetic Showers

Electrons, positrons, and photons initiate electromagnetic showers, which appear as diffuse, often conically expanding structures composed of many short, scattered track segments resulting from pair production and bremsstrahlung.

Beyond these primary topologies, two important secondary structures provide additional physics information:

- **Michel electrons** arise from the decay of a stopping muon ($\mu \to e^- \bar{\nu}\nu$) and manifest as short (typically few cm), sometimes slightly curved electron tracks originating specifically from the Bragg peak of a muon track.
- **Delta rays** are energetic electrons knocked out of Ar atoms by a high-energy charged particle traversing the medium; they appear as short electron tracks branching off directly from the side of a primary particle track.

Motivation: From Supervised to Self-Supervised Learning

Current State-of-the-Art & Its Trade-offs

The current state-of-the-art approach for LArTPC data reconstruction leverages deep neural networks trained with supervised learning on large, detailed Monte Carlo simulated datasets, as demonstrated by frameworks such as SPINE (Drielsma et al., 2021b). This "simulate → supervised train → calibrate" paradigm has proven highly effective and is central to ongoing analyses. However, it involves certain trade-offs:

- Significant investment in simulation generation and careful calibration to mitigate potential discrepancies between simulation and real detector data (**domain shift**)
- Models trained for one detector geometry or condition may not be directly transferable
- Managing distinct networks for the multitude of reconstruction tasks can be complex
- Computational resources required for large-scale simulation remain a consideration for next-generation experiments

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Self-Supervised Learning as a Promising Path Forward

SSL learns representations from the inherent structure of unlabeled data itself. In the **"pre-train → fine-tune"** paradigm, a large model is initially pre-trained on a vast corpora of unlabeled data to learn efficient representations, then fine-tuned in a supervised manner to adapt to any downstream task with a small amount of data — often described as **'few-shot learning'**.

Main Contributions

1

First SSL for LArTPCs

The first successful application and adaptation of self-supervised masked modeling for representation learning directly on sparse 3D point cloud data from LArTPCs.

2

Physically Meaningful Representations

Demonstration that SSL-pretrained representations capture physically meaningful structures, evidenced by emergent instance segmentation in attention maps and extreme data efficiency for track/shower classification.

3

C-NMS Tokenization

Introduction and validation of C-NMS, a volumetric tokenization strategy tailored for sparse particle trajectory data, and demonstration of the benefit of an auxiliary energy prediction task.

4

PILArNet-M Dataset

The public release of PILArNet-M, a large LArTPC simulation dataset, to serve as a benchmark and resource for future ML research in particle physics.

Dataset and Evaluation

The dataset comprises **1,210,080 simulated LArTPC images** generated using standard HEP simulation tools. Each image represents particle interactions within a cubic detector volume of $(2.3\ \mathrm{m})^3$, discretized into 768^3 voxels with a 3 mm pitch.

Metadata Labels

Fragment ID	A unique identifier (per event) for the smallest contiguous clusters of energy depositions, useful for low-level clustering tasks.
Group ID	A unique identifier (per event) aggregating fragments belonging to the same physical particle trajectory (e.g., a single muon track), often corresponding to "particles" in physics analyses.
Interaction ID	A unique identifier (per event) grouping particles originating from the same primary physics interaction vertex.
Semantic Type	An integer classifying the particle type: electromagnetic showers (0), MIP tracks (1), Michel electrons (2), delta rays (3), and low-energy background (4).

Data Preprocessing

- Voxels labeled as low-energy depositions (ID: 4) are removed, as these typically represent noise or diffuse background hits.
- Only events containing at least 1024 remaining points after this cut are considered.
- 3D coordinates (xi) are normalized to lie within a unit sphere centered at the origin.
- Energy depositions (Ei) are log-transformed to handle their large dynamic range and then scaled to the interval $[-1, 1]$.

Training Setup & Evaluation Metrics

Training and Validation Sets

1.03M

Training Events

Full dataset meeting preprocessing criteria, used for self-supervised pre-training

10.5K

Validation Events

Held-out set reserved for validation and testing

30

Linear Probe Events

Small random subset for evaluating pre-trained representations

For evaluating the downstream semantic segmentation task after fine-tuning, we train on subsets of the main training set (with varying sizes, down to 100 events) and evaluate performance on the full 10,471-event held-out validation set.

Evaluation Metrics

We employ **two stages of evaluation**: one during pre-training, and one during fine-tuning.

Pre-training evaluation: We train linear Support Vector Machines (SVMs) on the frozen representations (tokens) produced by the pre-trained encoder. Since each token represents a group of points that may contain multiple semantic types, we train separate One-vs-Rest (OvR) classifiers to predict the presence of each semantic type within a token. Performance is measured using the **F1-score** for each class.

Fine-tuning evaluation: After fine-tuning a segmentation head on top of the pre-trained encoder (using varying amounts of labeled data), we evaluate point-wise semantic segmentation performance. We report standard classification metrics, including per-class F1-scores and precision, allowing direct comparison with the fully supervised Sparse UResNet baseline.

Method: PoLAr-MAE Architecture

Overall Architecture and Pre-training Strategy

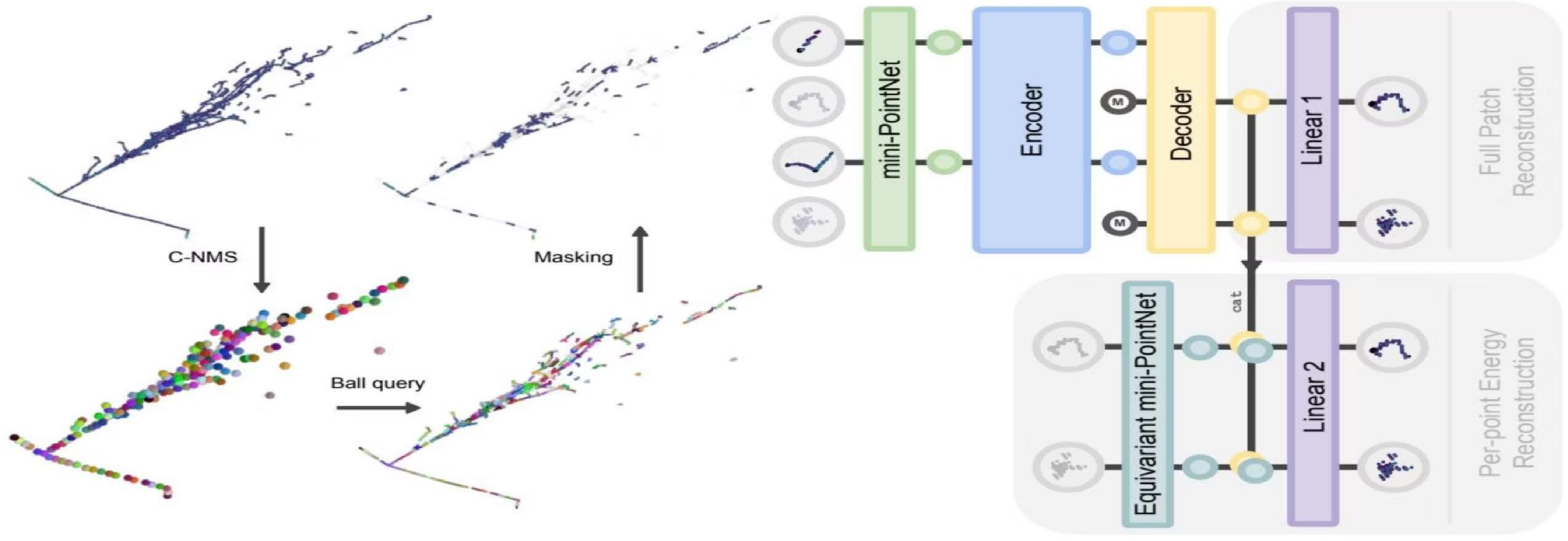


Figure 2 outlines PoLAr-MAE's overall architecture and pretraining strategy, adapted from Point-MAE with an added energy reconstruction task.

The LArTPC image is converted into a point cloud and processed through a modified grouping module. Some groups are masked, while visible ones are embedded into latent discrete tokens. A **Vision Transformer (ViT)**-based autoencoder, with a heavy (many parameters) encoder and light (comparatively fewer parameters) decoder, reconstructs full patches.

For energy reconstruction, a permutation-equivariant embedding module processes 3D point positions, and a linear head predicts per-point energy using decoded masked embeddings and encoded 3D positions.

Tokenization of Particle Trajectories

Patch Grouping with C-NMS

The common method for grouping point clouds into smaller patches involves farthest point sampling (FPS) followed by k-NN or ball query. However, in our dataset where point density varies, traditional FPS combined with k-NN or ball queries proves insufficient — resulting in either excessive ungrouped points or excessive overlap between local patches (see Figure 3).

To address this, we introduce **Centrality-based Non-Maximum Suppression (C-NMS)**, a volumetric sampling method extending conventional NMS to operate on spherical regions.

01

Sample Candidate Centers

A large set of candidate group centers is sampled, e.g., via FPS or by taking the entire point cloud.

02

Define Spheres

Each group center is treated as the center of a sphere with a fixed radius r.

03

Greedy NMS

Overlapping spheres are iteratively removed until no two remaining spheres exceed a predefined overlap factor f.

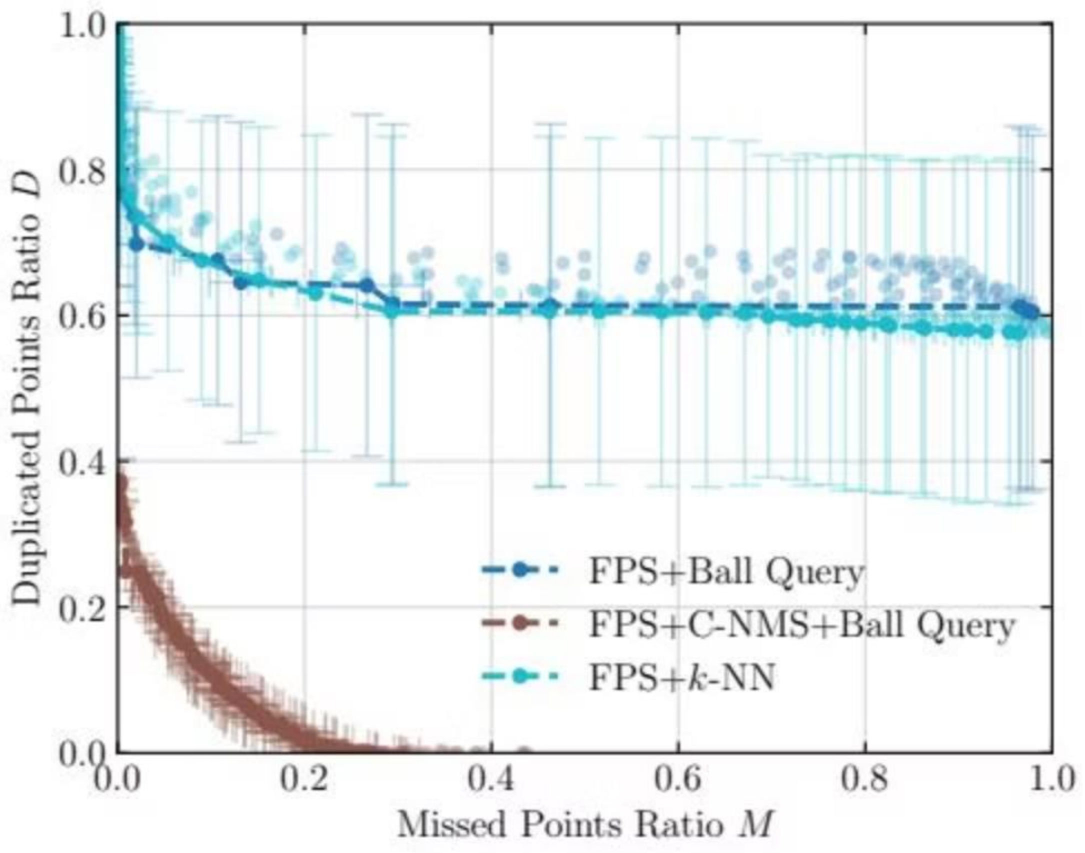
04

Ball Query Grouping

Points for each group are sampled via a ball query using the same radius r, forming G groups of variable size K.

Unlike FPS+k-NN, which lacks control over patch overlap, C-NMS ensures a tunable overlap between patches. C-NMS dynamically determines both the number of tokens G and the number of points per token K based on two parameters: the sphere radius r and the overlap factor f. This method has been implemented as a custom CUDA kernel for extremely fast processing.

After grouping, each patch is normalized by subtracting the patch center: $x_i \rightarrow x_i - c$, and scaled by the sphere radius r so that all patch points' positional coordinates lie within $[-1, 1]^3$. Energy values are not normalized.



Patch Encoding & Masked Event Modeling

Patch Encoding

Each patch group is encoded into a single latent vector using a **permutation-invariant mini-PointNet** — a small MLP that uplifts each point to a higher latent dimension and performs max-pooling to collapse all point embeddings into a single vector. Groups with fewer than K points are padded with copies of the first point.

Patch embeddings form the input sequence to a **vanilla Transformer encoder**, which computes global contextual information across all patch tokens. Positional embeddings are learned via a three-layer MLP mapping 3D patch center coordinates to latent vectors.

Masked Event Modeling

A masking module selects a random subset of patch tokens. Unmasked tokens are processed through the encoder, producing contextually enriched embeddings. A shallow decoder takes encoded visible tokens along with learnable mask tokens, with positional embeddings added to both.

For each masked token, the decoder outputs predicted points. Reconstruction is evaluated using the **Chamfer Distance loss**:

$$d_{CD}(S_1, S_2) = \sum_{x \in S_1} \min_{y \in S_2} ||x - y||_2^2 + \sum_{y \in S_2} \min_{x \in S_1} ||x - y||_2^2.$$

Auxiliary Energy Reconstruction

We introduce an auxiliary energy reconstruction task, whereby the network learns to predict per-point energies given their individual 3D positions. This is a natural extension, as the energy deposition over the length of a given trajectory (dE/dx) is an important discriminator for particle identification (PID) in any LArTPC analysis.

Pre-training Configuration

Hardware

4× NVIDIA A100 GPUs

Iterations

500,000 (~60 epochs)

Batch Size

Effective batch size of 512

Masking Ratio

60% of input tokens

Key Hyperparameters

Grouping Radius	5 voxels (15 mm)
C-NMS Overlap Fraction (f)	0.73 (optimized)
Max Points per Group	32 (FPS applied when exceeding)
Encoder/Decoder	ViT-S-like specifications
Avg. Groups per Event	~400

Quantitative Results

Linear Token Classification

concise quantitative summary of linear token classification results and implications for downstream particle ID.

Key takeaways

- Model learns semantic particle-level features without explicit particle ID labels.
- Tracks: SVM on pretrained representations achieves 99.4% F1.
- Showers: SVM on pretrained representations achieves 97.7% F1.

Quick context

The linear probe (SVM) was applied to token representations after pretraining. High F-scores show semantic structure emerges rapidly and is linearly separable.

Fine-tuning Approach

We fine-tune the PoLAr-MAE encoder for dense voxel-wise classification into the four particle types above. This section presents segmentation results that bridge representation learning to downstream performance.

Fine-tuning strategies

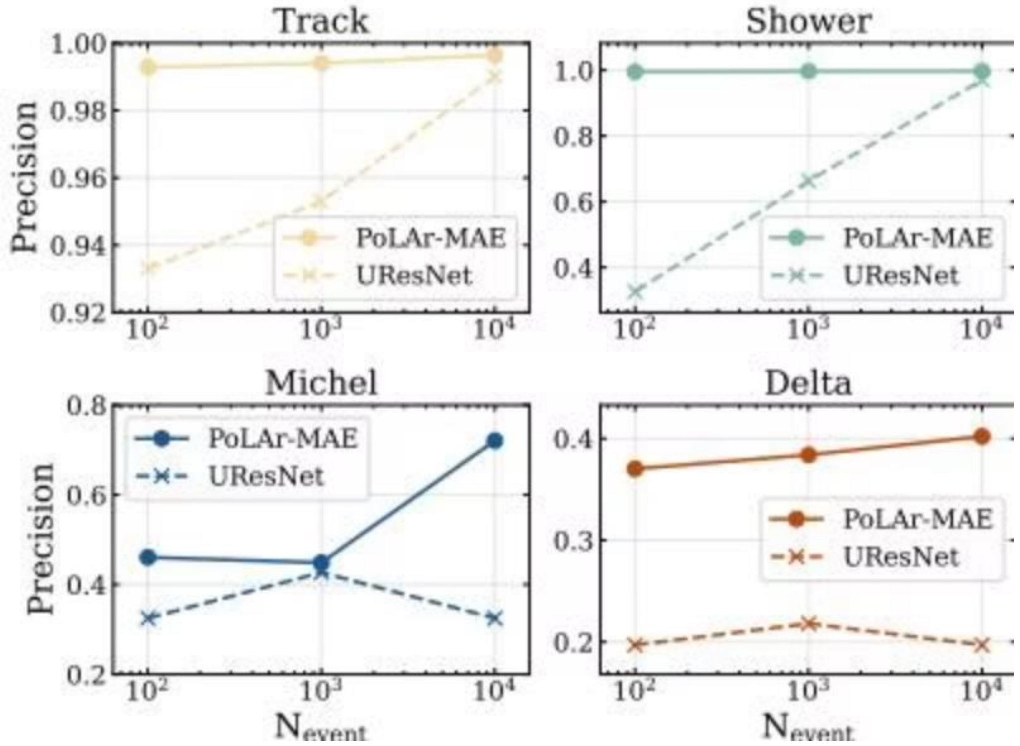
- **Full Fine-Tuning**: optimize all model parameters (encoder + heads)
- **Parameter-Efficient Fine-Tuning (PEFT)**: freeze encoder; train only upsampling & segmentation heads

Data scales

1. 100 events (low-data)
2. 1,000 events (medium-data)
3. 10,000 events (higher-data)

Baseline for comparison

UResNet (supervised) — fully supervised baseline trained on the full dataset used in prior work.



Comparison to Supervised Approach

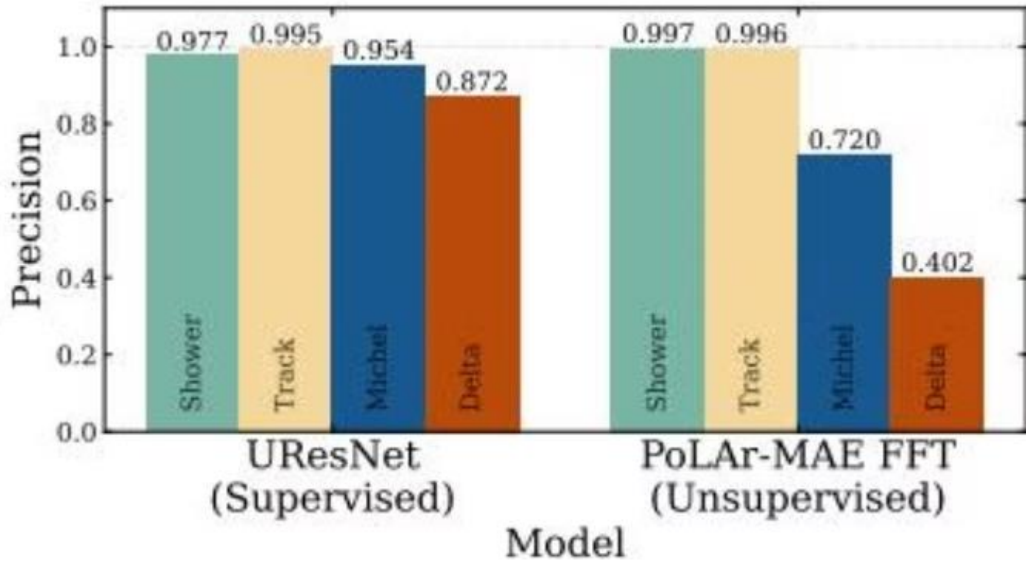


Figure 7: PoLAr-MAE fine-tuned on 10,000 events vs. SPINE trained on 100,000 events. Our model outperforms on track and shower classification while training on 10× less labeled data.

Qualitative Analysis: PCA Visualization

This PCA-based visualization maps each token's latent embedding to a color so we can see what the model has learned in an intuitive way. Each token (the center of a patch) is projected into RGB using the top principal components of its embedding space, producing a colored point for every patch in an event.

Important preprocessing: we remove spatial bias before coloring by regressing out the effect of patch center coordinates from the embeddings. This ensures the colors reflect the learned semantic content of the representations, not just where the patch sits in the detector.

What the visualization shows, in simple terms:

- Colors represent latent vector embeddings — similar colors mean similar embeddings.
- After removing spatial bias, color differences indicate semantic differences learned by the model, not positional artifacts.
- The pretrained models clearly separate particle types: tracks, showers, Michel electrons, and delta-rays tend to occupy distinct color regions.

Key insight: Tokens along the same particle trajectory (or from the same physical vertex) are colored similarly, showing that similar trajectories have similar latent representations; conversely, different particle types show different colors, indicating the model captures meaningful semantic distinctions.

This qualitative result complements the quantitative metrics and serves as a bridge to the next section on attention maps: together, the PCA colors and attention visualizations reveal both what the model encodes for individual patches and how it distributes focus across a full event.

PCA of Patch Features

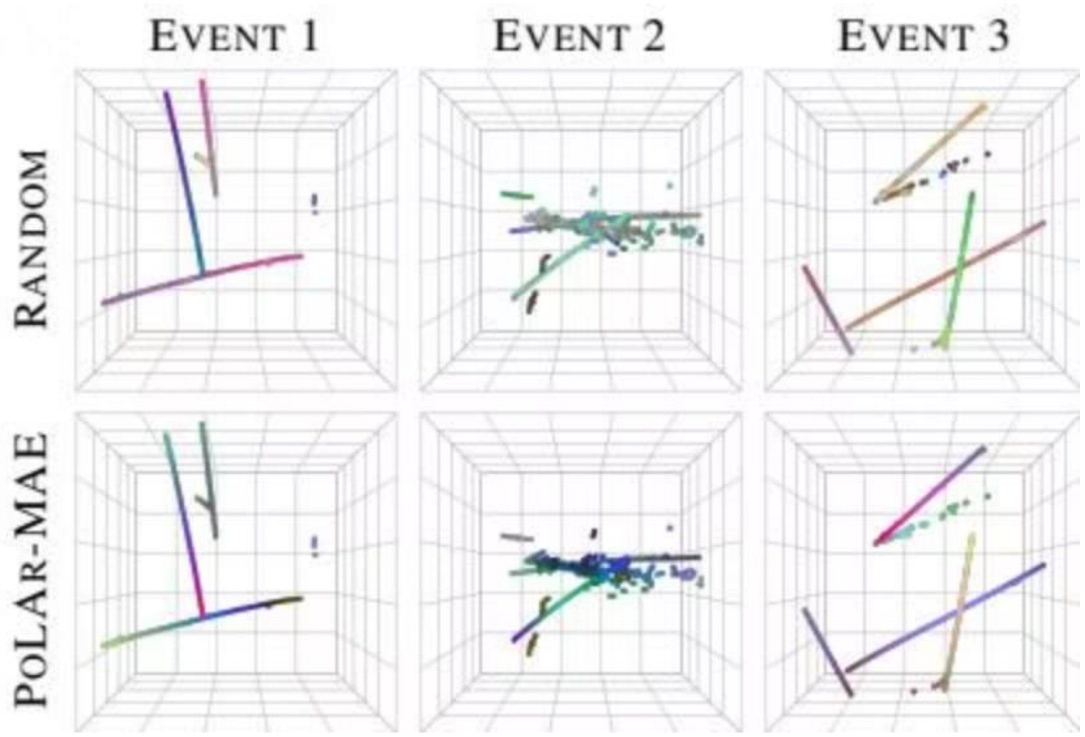


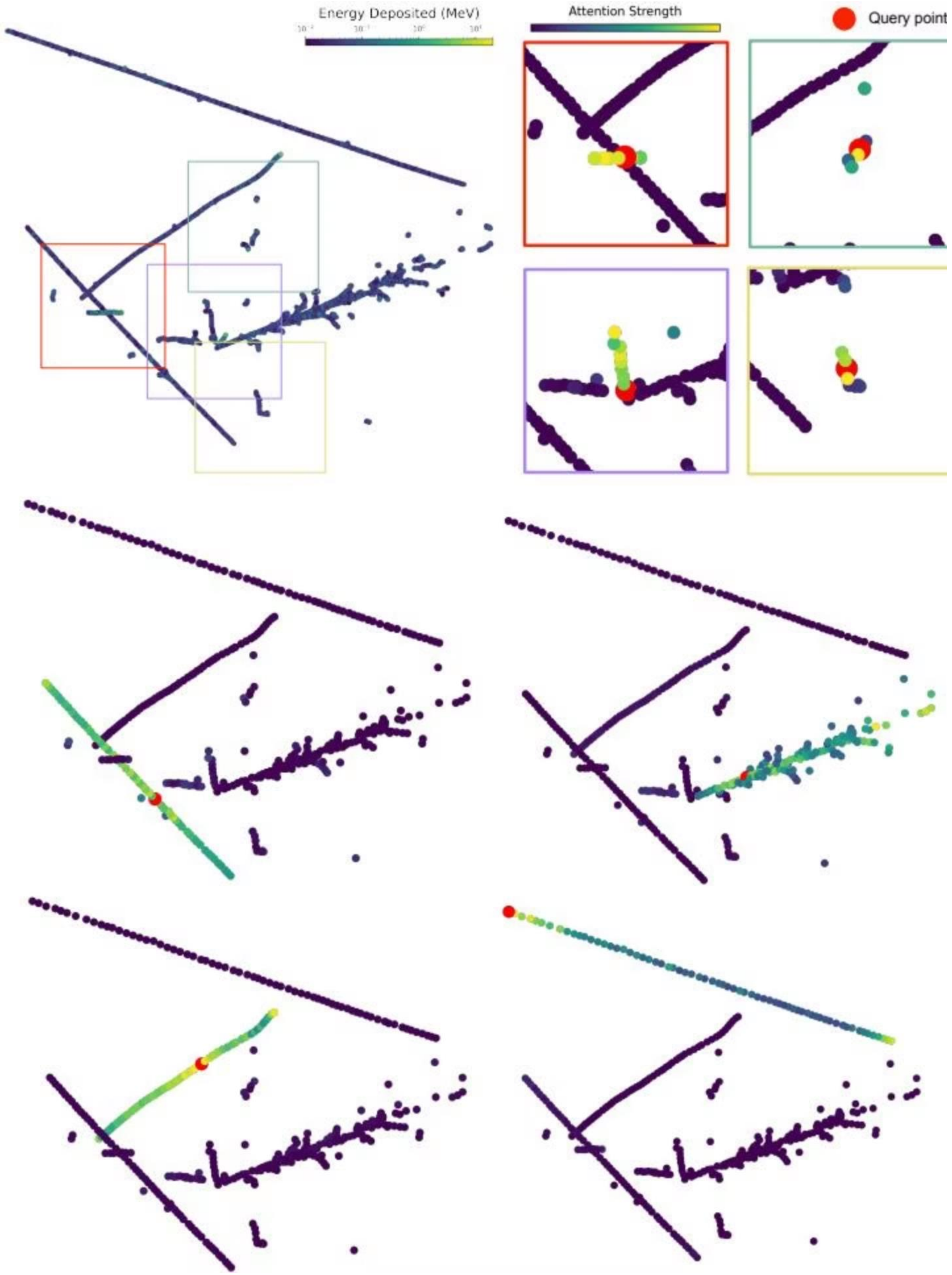
Figure 8: Spatially debiased learned representations. The randomly initialized model displays a strong positional bias, while PoLAr-MAE shows a clearer bias towards individual trajectories.

Qualitative Results: Emergent Representations

Attention Maps: Emergent Instance Segmentation

We visualize the Transformer encoder's attention to see how the model connects different parts of an event. In each plot a single token is chosen as the **query point** (shown in red), and every other token is colored by how much attention it gives to that query. Bright yellow means high attention — those tokens are the most relevant to the query point — while darker colors mean less attention.

The visualizations consistently show that attention is not random: tokens that lie along the same particle trajectory (for example, a track or the spread of an electromagnetic shower) tend to receive strong attention from a queried token on that trajectory. In other words, when the model looks at one part of a particle, it often focuses on other parts of the same particle.



Future Work and Discussion

This work presents the first successful implementation of a self-supervised, masked autoencoding framework for representation learning directly on sparse 3D point cloud data from LArTPCs. The high F_1 scores for track (0.994) and shower (0.977) token classification validate the efficacy of our SSL approach. These metrics are on par with state-of-the-art supervised frameworks trained on >100,000 events, despite not relying on any labeled examples.

However, the suboptimal modeling of highly-infrequent particles such as Michel electrons and delta rays indicates a limitation — highlighting the **"sim2real"** gap challenge and the need for architectural innovations.

Hierarchical Architectures

TPC data is inherently multi-scale. Hierarchical designs (e.g., Hiera backbone in SAM 2) enable more efficient parameter use and robust multi-scale feature learning, starting spatially fine-grained and becoming spatially coarser yet richer in features.

Point–Native Architectures

Point Transformers and Erwin offer a promising direction by operating directly on individual point features, addressing limitations of fixed-resolution token-based ViT-like models.

Alternative SSL Paradigms

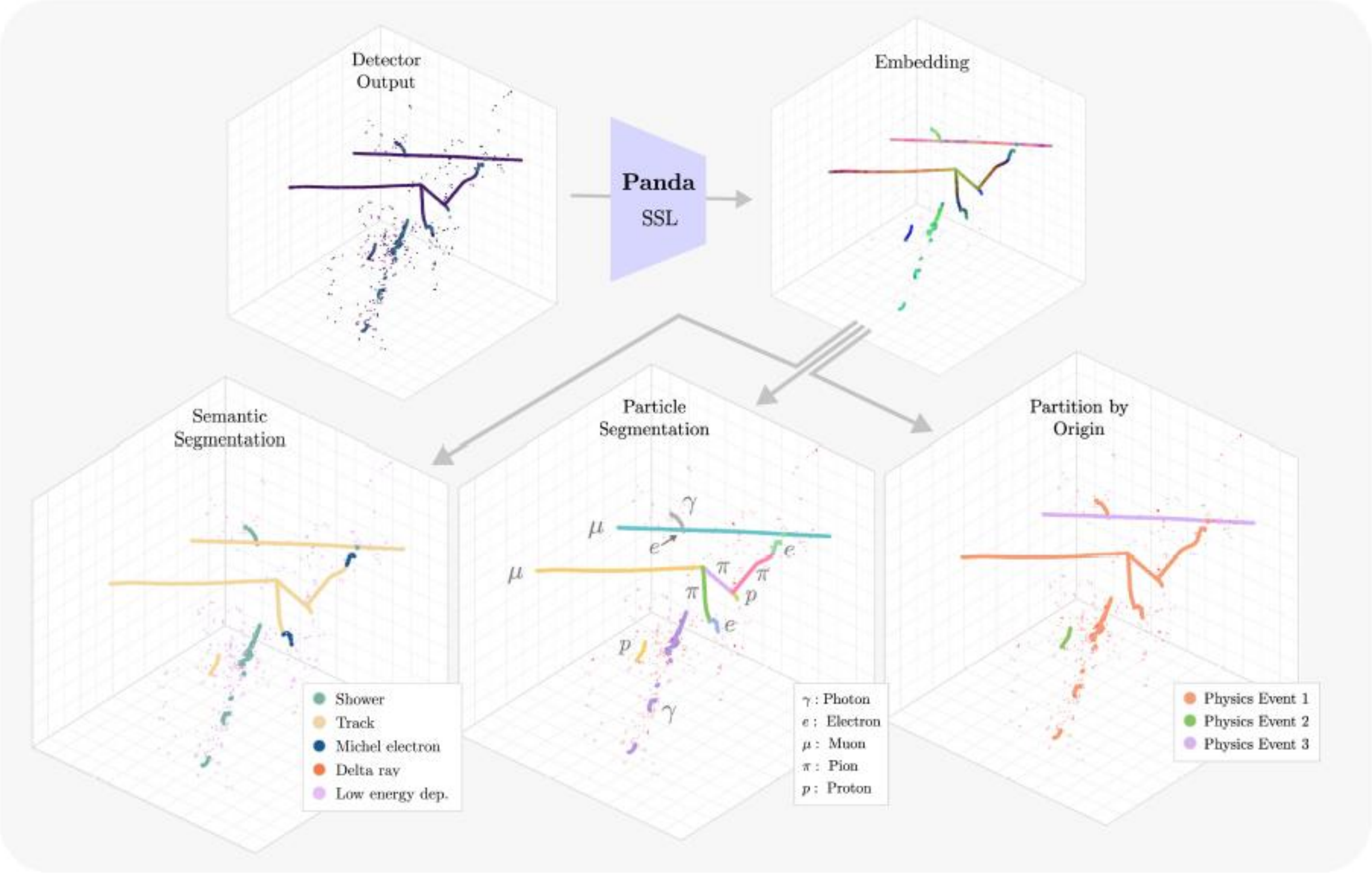
Contrastive learning, student-teacher self-distillation for view consistency, masked latent prediction, and local-global view consistency may also offer benefits and should be explored.

We hope that the introduction of the PILArNet-M dataset into the community will spur additional research in the quest for creating a LArTPC **foundation model**.

Panda Framework Overview

A single pre-trained backbone supports **all downstream tasks** without detector-specific heuristics.

Raw 3D charge depositions are encoded via a **point-native hierarchical encoder pre-trained via self-distillation**, producing shared embeddings reused for semantic segmentation, particle-level panoptic segmentation, and interaction-level partitioning — all with lightweight heads.



Thank you for your attention.